

Formula Sheet

ELG 5218 Uncertainty Evaluation in
Engineering Measurements and Machine
Learning

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1 Core Bayesian Identities

1.1 Bayes' Theorem

$$p(\theta | D) = \frac{p(D | \theta) p(\theta)}{p(D)}, \quad (1)$$

$$p(D) = \int p(D | \theta) p(\theta) d\theta. \quad (2)$$

Posterior is often written up to proportionality:

$$p(\theta | D) \propto p(D | \theta) p(\theta). \quad (3)$$

1.2 Likelihood for IID Data

For IID data $D = \{x_i\}_{i=1}^N$:

$$p(D | \theta) = \prod_{i=1}^N p(x_i | \theta), \quad (4)$$

$$\log p(D | \theta) = \sum_{i=1}^N \log p(x_i | \theta). \quad (5)$$

1.3 MLE and MAP

Maximum Likelihood Estimate (MLE):

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \log p(D | \theta). \quad (6)$$

Maximum A Posteriori (MAP):

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \log p(D | \theta) + \log p(\theta). \quad (7)$$

1.4 Posterior Predictive Distribution

$$p(x_* | D) = \int p(x_* | \theta) p(\theta | D) d\theta. \quad (8)$$

A plug-in (approximate) predictive:

$$p(x_* | D) \approx p(x_* | \hat{\theta}). \quad (9)$$

1.5 Uncertainty Quantification and Intervals

1.5.1 Credible Interval (Equal-Tailed)

For parameter θ with posterior $p(\theta | D)$, a $(1 - \alpha)$ central credible interval is

$$\text{CrI}_{1-\alpha} = [q_{\alpha/2}, q_{1-\alpha/2}], \quad (10)$$

where q_p is the p -quantile of $p(\theta | D)$.

1.5.2 Highest Posterior Density (HPD) Region

HPD region H with mass $(1 - \alpha)$ satisfies

$$\int_H p(\theta | D) d\theta = 1 - \alpha, \quad (11)$$

$$p(\theta | D) \geq c, \quad \forall \theta \in H, \quad (12)$$

for some threshold c chosen to achieve mass $1 - \alpha$.

2 Beta–Binomial Model

2.1 Model

$$k | \theta \sim \text{Binomial}(n, \theta), \quad (13)$$

$$\theta \sim \text{Beta}(\alpha, \beta). \quad (14)$$

2.2 Posterior

$$\theta | k, n \sim \text{Beta}(\alpha + k, \beta + n - k). \quad (15)$$

2.3 Moments and MAP

Posterior mean:

$$\mathbb{E}[\theta | k, n] = \frac{\alpha + k}{\alpha + \beta + n}. \quad (16)$$

Posterior variance:

$$\text{Var}[\theta | k, n] = \frac{(\alpha + k)(\beta + n - k)}{(\alpha + \beta + n)^2(\alpha + \beta + n + 1)}. \quad (17)$$

MAP estimator (for $\alpha, \beta > 1$):

$$\hat{\theta}_{\text{MAP}} = \frac{k + \alpha - 1}{n + \alpha + \beta - 2}. \quad (18)$$

3 Gaussian Distributions and Identities

3.1 Scalar Gaussian

$$p(x) = \mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right). \quad (19)$$

Precision form with $\lambda = 1/\sigma^2$:

$$p(x) \propto \sqrt{\lambda} \exp\left(-\frac{\lambda}{2}(x - \mu)^2\right). \quad (20)$$

3.2 Multivariate Gaussian

$$p(z) = \mathcal{N}(z; \mu, \Sigma) \quad (21)$$

$$= \frac{1}{(2\pi)^{D/2} \sqrt{\det \Sigma}} \exp\left(-\frac{1}{2}(z - \mu)^\top \Sigma^{-1}(z - \mu)\right). \quad (22)$$

Precision form with $\Lambda = \Sigma^{-1}$:

$$p(z) \propto |\Lambda|^{1/2} \exp\left(-\frac{1}{2}(z - \mu)^\top \Lambda (z - \mu)\right). \quad (23)$$

3.3 Partitioned Gaussian: Marginal and Conditional

Joint Gaussian:

$$\begin{bmatrix} x \\ y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{bmatrix}\right). \quad (24)$$

Marginal of x :

$$x \sim \mathcal{N}(\mu_x, \Sigma_{xx}). \quad (25)$$

3.4 Theorem 2: Conditional of x given y

$$p(x | y) = \mathcal{N}(x; \mu_{x|y}, \Sigma_{x|y}), \quad (26)$$

$$\mu_{x|y} = \mu_x + \Sigma_{xy} \Sigma_{yy}^{-1} (y - \mu_y), \quad (27)$$

$$\Sigma_{x|y} = \Sigma_{xx} - \Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx}. \quad (28)$$

This is Theorem 2 (conditioning) expressed with (x, y) .

3.5 Theorem 3: Affine Transformation

Assume

$$p(x) = \mathcal{N}(x; \mu_x, \Sigma_x), \quad (29)$$

$$p(y | x) = \mathcal{N}(y; Ax + b, \Sigma_{y|x}), \quad (30)$$

where the conditional $y | x$ is a linear function of x plus Gaussian noise. Then the stacked vector $(x, y)^\top$ is jointly Gaussian:

$$p(x, y) = \mathcal{N}\left(\begin{pmatrix} x \\ y \end{pmatrix}; \begin{pmatrix} \mu_x \\ A\mu_x + b \end{pmatrix}, R\right), \quad (31)$$

with block covariance

$$R = \begin{pmatrix} \Sigma_x & \Sigma_x A^\top \\ A \Sigma_x & \Sigma_{y|x} + A \Sigma_x A^\top \end{pmatrix}. \quad (32)$$

Corollary 2: Marginal of y (Predictive Distribution)

$$p(y) = \int p(y | x) p(x) dx = \mathcal{N}(y; \mu_y, \Sigma_y), \quad (33)$$

where

$$\mu_y = A\mu_x + b, \quad (34)$$

$$\Sigma_y = \Sigma_{y|x} + A \Sigma_x A^\top. \quad (35)$$

Corollary 1: Posterior of x given y

$p(x | y)$:

$$p(x | y) = \mathcal{N}(x; \mu_{x|y}, \Sigma_{x|y}), \quad (36)$$

with

$$\Sigma_{x|y} = (\Sigma_x^{-1} + A^\top \Sigma_{y|x}^{-1} A)^{-1}, \quad (37)$$

$$\mu_{x|y} = \Sigma_{x|y} (\Sigma_x^{-1} \mu_x + A^\top \Sigma_{y|x}^{-1} (y - b)). \quad (38)$$

3.6 Product and Ratio of Scalar Gaussians

Product: if

$$\mu_1(x) \propto \mathcal{N}(x; m_1, \sigma_1^2), \quad (39)$$

$$\mu_2(x) \propto \mathcal{N}(x; m_2, \sigma_2^2), \quad (40)$$

then

$$\mu_1(x) \mu_2(x) \propto \mathcal{N}(x; m, \sigma^2), \quad (41)$$

$$\sigma^2 = \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} \right)^{-1}, \quad (42)$$

$$m = \sigma^2 \left(\frac{m_1}{\sigma_1^2} + \frac{m_2}{\sigma_2^2} \right). \quad (43)$$

Ratio: with same μ_1, μ_2 ,

$$\frac{\mu_1(x)}{\mu_2(x)} \propto \mathcal{N}(x; \tilde{m}, \tilde{\sigma}^2), \quad (44)$$

$$\tilde{\sigma}^2 = \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2} \right)^{-1}, \quad (45)$$

$$\tilde{m} = \tilde{\sigma}^2 \left(\frac{m_1}{\sigma_1^2} - \frac{m_2}{\sigma_2^2} \right). \quad (46)$$

4 Gaussian Models and Conjugate Updates

4.1 4.1 Known Variance, Unknown Mean (Scalar)

Model: $x_i | \mu, \sigma^2$ i.i.d. $\mathcal{N}(\mu, \sigma^2)$, known σ^2 , precision $\lambda = 1/\sigma^2$.

Uniform Prior $p(\mu) \propto 1$:

$$\mu | x_{1:n} \sim \mathcal{N}\left(\bar{x}, \frac{\sigma^2}{n}\right), \quad (47)$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.

Normal Prior $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$ with precision $\lambda_0 = 1/\sigma_0^2$:

$$\lambda_n = \lambda_0 + n\lambda, \quad (48)$$

$$\mu_n = \frac{\lambda_0 \mu_0 + n\lambda \bar{x}}{\lambda_n}, \quad (49)$$

$$\mu | x_{1:n} \sim \mathcal{N}\left(\mu_n, \frac{1}{\lambda_n}\right). \quad (50)$$

4.2 Sequential (Online) Update for Mean

Let μ_{n-1}, λ_{n-1} be posterior parameters after $n-1$ points; add x_n .

$$\lambda_n = \lambda_{n-1} + \lambda, \quad (51)$$

$$w_n = \frac{\lambda}{\lambda_n}, \quad (52)$$

$$\mu_n = \mu_{n-1} + w_n(x_n - \mu_{n-1}). \quad (53)$$

4.3 Unknown Mean and Variance: Normal-Gamma Prior

Use precision $\lambda = 1/\sigma^2$.

$$\mu | \lambda \sim \mathcal{N}(\mu_0, (\kappa_0 \lambda)^{-1}), \quad (54)$$

$$\lambda \sim \text{Gamma}(\alpha_0, \beta_0). \quad (55)$$

Posterior hyperparameters after n observations with sample mean $\bar{x}$:

$$\kappa_n = \kappa_0 + n, \quad (56)$$

$$\mu_n = \frac{\kappa_0 \mu_0 + n \bar{x}}{\kappa_n}, \quad (57)$$

$$\alpha_n = \alpha_0 + \frac{n}{2}, \quad (58)$$

$$\beta_n = \beta_0 + \frac{1}{2} \left(\sum_{i=1}^n (x_i - \bar{x})^2 + \frac{\kappa_0 n}{\kappa_0 + n} (\bar{x} - \mu_0)^2 \right). \quad (59)$$

Posterior:

$$p(\mu, \lambda \mid x_{1:n}) = \mathcal{N}(\mu; \mu_n, (\kappa_n \lambda)^{-1}) \text{Gamma}(\lambda; \alpha_n, \beta_n). \quad (60)$$

4.4 Student-t Distribution from Normal-Gamma

If μ, λ have Normal-Gamma posterior as above, the marginal for μ is Student- t :

$$\mu \mid x_{1:n} \sim t_{2\alpha_n} \left(\mu_n, \sqrt{\frac{\beta_n}{\alpha_n \kappa_n}} \right). \quad (61)$$

Posterior predictive (new x_*):

$$x_* \mid x_{1:n} \sim t_{2\alpha_n} \left(\mu_n, \sqrt{\frac{\beta_n}{\alpha_n} \left(1 + \frac{1}{\kappa_n} \right)} \right). \quad (62)$$

5 Bayesian Linear Regression

5.1 Linear Regression Model

For real-valued targets $y_n \in \mathbb{R}$ and feature vectors x_n :

$$y_n \mid x_n, w \sim \mathcal{N}(w^\top x_n, \sigma^2), \quad (63)$$

$$p(y_n \mid x_n, w) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} (y_n - w^\top x_n)^2\right). \quad (64)$$

Stack all data:

$$y = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}, \quad X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_N^\top \end{bmatrix}.$$

5.2 Likelihood

$$p(y \mid X, w) = \prod_{n=1}^N \mathcal{N}(y_n \mid w^\top x_n, \sigma^2) \quad (65)$$

$$= (2\pi\sigma^2)^{-N/2} \exp\left(-\frac{1}{2\sigma^2} \|y - Xw\|^2\right). \quad (66)$$

5.3 Gaussian Prior on Weights

$$p(w) = \mathcal{N}(w; 0, \lambda^{-1} I), \quad (67)$$

$$\log p(w) = -\frac{\lambda}{2} w^\top w + C. \quad (68)$$

5.4 Posterior (Up to Proportionality)

$$p(w \mid X, y) \propto \exp\left(-\frac{1}{2\sigma^2} \|y - Xw\|^2 - \frac{\lambda}{2} w^\top w\right). \quad (69)$$

5.5 Closed-Form Gaussian Posterior

Posterior precision and covariance:

$$\Sigma_N^{-1} = \lambda I + \frac{1}{\sigma^2} X^\top X, \quad (70)$$

$$\Sigma_N = \left(\lambda I + \frac{1}{\sigma^2} X^\top X \right)^{-1}. \quad (71)$$

Posterior mean:

$$\mu_N = \Sigma_N \left(\frac{1}{\sigma^2} X^\top y \right). \quad (72)$$

Thus the posterior is:

$$p(w \mid D) = \mathcal{N}(w; \mu_N, \Sigma_N). \quad (73)$$

MAP estimate:

$$w_{\text{MAP}} = \mu_N. \quad (74)$$

5.6 Posterior Predictive Distribution

For a new point x_* :

$$p(y_* \mid x_*, D) = \int \mathcal{N}(y_* \mid w^\top x_*, \sigma^2) \mathcal{N}(w; \mu_N, \Sigma_N) dw. \quad (75)$$

Closed form:

$$y_* \mid x_*, D \sim \mathcal{N}(\mu_N^\top x_*, \sigma^2 + x_*^\top \Sigma_N x_*). \quad (76)$$

Predictive mean:

$$\mu_* = \mu_N^\top x_*.$$

Predictive variance:

$$\text{Var}(y_*) = \sigma^2 + x_*^\top \Sigma_N x_*.$$

6 Logistic Regression and Laplace Approximation

6.1 Logistic Model

For binary labels $y_n \in \{0, 1\}$ and features x_n :

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad (77)$$

$$p(y_n = 1 \mid x_n, w) = \sigma(w^\top x_n), \quad (78)$$

$$p(y_n = 0 \mid x_n, w) = 1 - \sigma(w^\top x_n). \quad (79)$$

Likelihood (Bernoulli factorization):

$$p(y \mid X, w) = \prod_{n=1}^N \sigma(w^\top x_n)^{y_n} (1 - \sigma(w^\top x_n))^{1-y_n}. \quad (80)$$

Gaussian prior on weights:

$$p(w) = \mathcal{N}(w; 0, \alpha^{-1} I). \quad (81)$$

Posterior (up to proportionality):

$$p(w \mid y, X) \propto p(y \mid X, w) p(w). \quad (82)$$

6.2 Log-Posterior, Gradient, Hessian

Let $\mu_n = \sigma(w^\top x_n)$ and define $\mu = (\mu_1, \dots, \mu_N)^\top$. Log-posterior (up to constant):

$$\log p(w | D) = -\frac{\alpha}{2} w^\top w + \sum_{n=1}^N \left[y_n \log \mu_n + (1 - y_n) \log(1 - \mu_n) \right] + C. \quad (83)$$

Gradient:

$$\nabla_w \log p(w | D) = -\alpha w + \sum_{n=1}^N (y_n - \mu_n) x_n \quad (84)$$

$$= -\alpha w + X^\top (y - \mu). \quad (85)$$

Hessian:

$$H = \nabla_w^2 \log p(w | D) \quad (86)$$

$$= -\alpha I - \sum_{n=1}^N \mu_n (1 - \mu_n) x_n x_n^\top \quad (87)$$

$$= -(\alpha I + X^\top S X), \quad (88)$$

where S is diagonal with $S_{nn} = \mu_n (1 - \mu_n)$.

6.3 Laplace Approximation for Posterior

Mode (MAP):

$$w_{\text{MAP}} = \arg \max_w \log p(w | D). \quad (89)$$

Quadratic (Laplace) approximation around w_{MAP} :

$$p(w | D) \approx \mathcal{N}(w; w_{\text{MAP}}, \Sigma_N), \quad (90)$$

$$\Sigma_N^{-1} = -H|_{w=w_{\text{MAP}}} = \alpha I + X^\top S X. \quad (91)$$

6.4 Predictive Probability (Approximate)

Exact predictive for new x_* :

$$p(y_* = 1 | x_*, D) = \int \sigma(w^\top x_*) p(w | D) dw. \quad (92)$$

Under Laplace approximation $p(w | D) \approx \mathcal{N}(w_{\text{MAP}}, \Sigma_N)$ this is a 1D Gaussian integral in $w^\top x_*$ (often evaluated numerically or via standard approximations).

6.5 Potential Energy for Logistic Regression

Define unnormalized posterior $\pi(w) \propto p(y | X, w) p(w)$. Potential energy:

$$U(w) = -\log \pi(w) \quad (93)$$

$$= -\sum_{n=1}^N [y_n \log \mu_n + (1 - y_n) \log(1 - \mu_n)] + \frac{\alpha}{2} w^\top w + C. \quad (94)$$

Gradient of $U(w)$:

$$\nabla U(w) = X^\top (\mu - y) + \alpha w. \quad (95)$$

7 Markov Chain Monte Carlo (MCMC)

7.1 Metropolis–Hastings Algorithm

Given target density $\pi(\theta)$ and proposal $q(\theta' | \theta)$:

$$r(\theta \rightarrow \theta') = \frac{\tilde{\pi}(\theta') q(\theta | \theta')}{\tilde{\pi}(\theta) q(\theta' | \theta)}, \quad (96)$$

$$\alpha(\theta \rightarrow \theta') = \min(1, r(\theta \rightarrow \theta')), \quad (97)$$

where $\tilde{\pi}$ is any unnormalized version of π . If $\pi(\theta) \propto p(x | \theta) p(\theta)$, then

$$r(\theta \rightarrow \theta') = \frac{p(x | \theta') p(\theta') q(\theta | \theta')}{p(x | \theta) p(\theta) q(\theta' | \theta)}, \quad (98)$$

so the evidence cancels.

7.2 Random-Walk Metropolis (RWM)

Symmetric proposal:

$$q(\theta' | \theta) = \mathcal{N}(\theta'; \theta, \sigma_{\text{prop}}^2 I), \quad (99)$$

so acceptance probability simplifies to

$$\alpha(\theta \rightarrow \theta') = \min\left(1, \frac{\pi(\theta')}{\pi(\theta)}\right). \quad (100)$$

7.3 Detailed Balance Condition

A sufficient condition for stationarity of π is

$$\pi(\theta) P(\theta' | \theta) = \pi(\theta') P(\theta | \theta') \quad (101)$$

for all θ, θ' . Metropolis–Hastings transition kernel satisfies detailed balance.

7.4 Monte Carlo Estimation

Given samples $\{\theta^{(s)}\}_{s=1}^S$ from $\pi(\theta)$, an expectation

$$\mathbb{E}_\pi[f(\theta)] = \int f(\theta) \pi(\theta) d\theta \quad (102)$$

can be approximated as

$$\mathbb{E}_\pi[f(\theta)] \approx \frac{1}{S} \sum_{s=1}^S f(\theta^{(s)}). \quad (103)$$

Standard error decays as $\sim 1/\sqrt{S}$ (modulo autocorrelation).

7.5 MCMC Diagnostics

7.5.1 Autocorrelation Function (ACF)

For an MCMC chain $\{\theta_t\}_{t=1}^N$, the autocorrelation at lag ℓ is:

$$\rho(\ell) = \text{Corr}(\theta_t, \theta_{t+\ell}). \quad (104)$$

7.5.2 Autocorrelation Time and Effective Sample Size (ESS)

Define the **autocorrelation time**:

$$\tau = 1 + 2 \sum_{\ell=1}^{\infty} \rho(\ell). \quad (105)$$

Effective sample size:

$$N_{\text{eff}} = \frac{N}{\tau}. \quad (106)$$

Monte Carlo standard error:

$$\text{SE}(\hat{\theta}) = \frac{\sigma(\theta)}{\sqrt{N_{\text{eff}}}}. \quad (107)$$

7.5.3 Gelman–Rubin Convergence Diagnostic ($\hat{R}$)

Suppose we run J parallel chains, each producing L post-burn-in samples.

Between-chain variance:

$$B = \frac{L}{J-1} \sum_{j=1}^J (\bar{\theta}_j - \bar{\theta})^2, \quad (108)$$

where $\bar{\theta}_j$ is the mean of chain j and $\bar{\theta}$ is the mean across chains.

Within-chain variance:

$$W = \frac{1}{J} \sum_{j=1}^J s_j^2, \quad (109)$$

where s_j^2 is the sample variance of chain j .

Potential Scale Reduction Factor (PSRF):

$$\hat{R} = \sqrt{\frac{(L-1)W + B}{LW}} = \sqrt{\frac{\text{between-chain variance}}{\text{within-chain variance}}}. \quad (110)$$

8 Langevin Dynamics and MALA

8.1 Langevin SDE

For a target density $p(x)$ with score $\nabla_x \log p(x)$, the Langevin stochastic differential equation is

$$dX_t = \frac{1}{2} \nabla_x \log p(X_t) dt + dW_t, \quad (111)$$

where W_t is standard Brownian motion.

8.2 Potential Form

For a target $\pi(\theta) \propto e^{-U(\theta)}$:

$$d\Theta_t = -\frac{1}{2} \nabla U(\Theta_t) dt + dW_t. \quad (112)$$

8.3 Unadjusted Langevin Algorithm (ULA)

Euler–Maruyama discretization with step size Δt :

$$\theta_{k+1} = \theta_k - \frac{1}{2} \nabla U(\theta_k) \Delta t + \sqrt{\Delta t} Z_k, \quad (113)$$

$$Z_k \sim \mathcal{N}(0, I). \quad (114)$$

8.4 Metropolis-Adjusted Langevin Algorithm (MALA)

Langevin proposal:

$$q(\tilde{\theta} | \theta) = \mathcal{N}\left(\tilde{\theta}; \theta - \frac{1}{2} \nabla U(\theta) \Delta t, \Delta t I\right). \quad (115)$$

Metropolis–Hastings acceptance probability:

$$\begin{aligned} \alpha(\theta \rightarrow \tilde{\theta}) &= \min\left(1, \frac{\pi(\tilde{\theta}) q(\theta | \tilde{\theta})}{\pi(\theta) q(\tilde{\theta} | \theta)}\right) \\ &= \min\left(1, \exp(-U(\tilde{\theta}) + U(\theta)) \frac{q(\theta | \tilde{\theta})}{q(\tilde{\theta} | \theta)}\right). \end{aligned} \quad (116)$$

$$(117)$$

9 Bayesian Filtering (General State-Space Model)

Density form: $p(z_k | z_{k-1}), \quad p(y_k | z_k).$

Prediction (Chapman–Kolmogorov):

$$p(z_k | y_{1:k-1}) = \int p(z_k | z_{k-1}) p(z_{k-1} | y_{1:k-1}) dz_{k-1}. \quad (118)$$

Update (Bayes' rule):

$$p(z_k | y_{1:k}) = \frac{p(y_k | z_k) p(z_k | y_{1:k-1})}{p(y_k | y_{1:k-1})}, \quad (119)$$

$$p(y_k | y_{1:k-1}) = \int p(y_k | z_k) p(z_k | y_{1:k-1}) dz_k. \quad (120)$$

Joint factorization (SSM):

$$p(z_{0:T}, y_{1:T}) = p(z_0) \prod_{k=1}^T p(z_k | z_{k-1}) \prod_{k=1}^T p(y_k | z_k). \quad (121)$$

9.1 Linear–Gaussian State Space Model (SSM)

Dynamics: $z_t = F_t z_{t-1} + B_t u_t + b_t + q_t, \quad q_t \sim \mathcal{N}(0, Q_t).$

$$p(z_t | z_{t-1}, u_t) = \mathcal{N}(z_t; F_t z_{t-1} + B_t u_t + b_t, Q_t). \quad (122)$$

Measurement: $y_t = H_t z_t + D_t u_t + d_t + r_t, \quad r_t \sim \mathcal{N}(0, R_t).$

$$p(y_t | z_t, u_t) = \mathcal{N}(y_t; H_t z_t + D_t u_t + d_t, R_t). \quad (123)$$

10 Kalman Filter (Complete Algorithm)

Initialization: $p(z_0) = \mathcal{N}(z_0; \mu_0, \Sigma_0).$ **For** $t=1, \dots, T$:

Prediction:

$$\begin{aligned} \mu_{t|t-1} &= F_t \mu_{t-1|t-1} + B_t u_t + b_t, \\ \Sigma_{t|t-1} &= F_t \Sigma_{t-1|t-1} F_t^\top + Q_t. \end{aligned} \quad (124)$$

Update:

$$\hat{y}_t = H_t \mu_{t|t-1} + D_t u_t + d_t, \quad S_t = H_t \Sigma_{t|t-1} H_t^\top + R_t, \quad (125)$$

$$K_t = \Sigma_{t|t-1} H_t^\top S_t^{-1}, \quad \mu_{t|t} = \mu_{t|t-1} + K_t(y_t - \hat{y}_t), \quad (126)$$

$$\Sigma_{t|t} = (I - K_t H_t) \Sigma_{t|t-1}. \quad (127)$$

Innovation (residual):
 $\tilde{y}_t = y_t - \hat{y}_t, \quad \tilde{y}_t \sim \mathcal{N}(0, S_t)$ (consistency check).

10.1 Kalman Filter (1D Scalar Special Case)

Model:

$$z_t = z_{t-1} + q_t, \quad q_t \sim \mathcal{N}(0, q_t), \quad y_t = z_t + r_t, \quad r_t \sim \mathcal{N}(0, r_t). \quad (128)$$

Prediction:

$$\mu_{t|t-1} = \mu_{t-1|t-1}, \quad \sigma_{t|t-1}^2 = \sigma_{t-1|t-1}^2 + q_t. \quad (129)$$

Update:

$$\hat{y}_t = \mu_{t|t-1}, \quad S_t = \sigma_{t|t-1}^2 + r_t, \quad K_t = \frac{\sigma_{t|t-1}^2}{\sigma_{t|t-1}^2 + r_t}, \quad (130)$$

$$\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - \hat{y}_t), \quad \sigma_{t|t}^2 = (1 - K_t) \sigma_{t|t-1}^2. \quad (131)$$

10.2 Monte Carlo and Importance Sampling

Monte Carlo expectation:

$$\mathbb{E}[f(Z)] = \int f(z) p(z) dz \approx \frac{1}{M} \sum_{m=1}^M f(z^{(m)}), \quad (132)$$

$$z^{(m)} \sim p(z). \quad (133)$$

Importance sampling identity:

$$\mathbb{E}[f(Z)] = \int f(z) \frac{p(z)}{q(z)} q(z) dz \approx \sum_{m=1}^M \tilde{w}^{(m)} f(z^{(m)}), \quad (134)$$

$$z^{(m)} \sim q(z), \quad (135)$$

$$w^{(m)} = \frac{p(z^{(m)})}{q(z^{(m)})}, \quad \tilde{w}^{(m)} = \frac{w^{(m)}}{\sum_{j=1}^M w^{(j)}}. \quad (135)$$

10.3 Sequential Importance Sampling / Particle Approximation

Empirical approximation:

$$p(z_k | y_{1:k}) \approx \sum_{m=1}^M w_k^{(m)} \delta(z_k - z_k^{(m)}). \quad (136)$$

General SMC weight update (observation-dependent proposal):

$$w_k^{(m)} \propto w_{k-1}^{(m)} \frac{p(y_k | z_k^{(m)}) p(z_k^{(m)} | z_{k-1}^{(m)})}{q(z_k^{(m)} | z_{k-1}^{(m)}, y_k)}. \quad (137)$$

Bootstrap / prior proposal ($q = p(z_k | z_{k-1})$):

$$w_k^{(m)} \propto w_{k-1}^{(m)} p(y_k | z_k^{(m)}). \quad (138)$$

10.4 Resampling (SIR / PF)

Resampling distribution: sample indices from

$$\sum_{j=1}^M w_k^{(j)} \delta(z_k - z_k^{(j)}), \quad (139)$$

then set new weights to $1/M$:

$$\{\bar{z}_k^{(m)}\}_{m=1}^M \sim \sum_{j=1}^M w_k^{(j)} \delta(z_k - z_k^{(j)}), \quad \bar{w}_k^{(m)} = \frac{1}{M}. \quad (140)$$

State estimate (posterior mean):

$$\hat{z}_k = \sum_{m=1}^M \bar{w}_k^{(m)} \bar{z}_k^{(m)}. \quad (141)$$

10.5 Optimal Proposal (Variance-Minimizing)

Optimal proposal:

$$q_k^{\text{opt}}(z_k | z_{k-1}, y_k) = p(z_k | z_{k-1}, y_k). \quad (142)$$

Weight simplification under q^{opt} :

$$w_k^{(m)} \propto w_{k-1}^{(m)} p(y_k | z_{k-1}^{(m)}), \quad (143)$$

$$p(y_k | z_{k-1}) = \int p(y_k | z_k) p(z_k | z_{k-1}) dz_k. \quad (144)$$

11 Optimization, Entropy, KL, and Variational Inference

11.1 Gradient Descent and SGD

For an objective $\mathcal{J}(\theta)$:

$$\theta_{t+1} = \theta_t - \eta_t \nabla_{\theta} \mathcal{J}(\theta_t). \quad (145)$$

Mini-batch stochastic gradient:

$$\widehat{\nabla_{\theta} \mathcal{J}(\theta)} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \nabla_{\theta} \ell_i(\theta). \quad (146)$$

11.2 Entropy and Differential Entropy

For a discrete random variable X :

$$H(X) = - \sum_i p_i \log p_i. \quad (147)$$

For a continuous density $p(x)$:

$$h(X) = - \int p(x) \log p(x) dx. \quad (148)$$

For $X \sim \mathcal{N}(\mu, \Sigma) \in \mathbb{R}^d$:

$$h(X) = \frac{1}{2} \log((2\pi e)^d |\Sigma|). \quad (149)$$

11.3 KL Divergence and Cross-Entropy

Discrete KL divergence:

$$\text{KL}(P\|Q) = \sum_i p_i \log \frac{p_i}{q_i}. \quad (150)$$

Continuous KL divergence:

$$\text{KL}(p\|q) = \int p(z) \log \frac{p(z)}{q(z)} dz. \quad (151)$$

Cross-entropy:

$$H(P, Q) = - \sum_i p_i \log q_i. \quad (152)$$

Relation:

$$\text{KL}(P\|Q) = H(P, Q) - H(P). \quad (153)$$

11.4 ELBO and Variational Gap

For latent variables z and variational density $q(z)$:

$$\mathcal{L}(q) = \mathbb{E}_{q(z)}[\log p(x, z)] - \mathbb{E}_{q(z)}[\log q(z)] \quad (154)$$

$$= \mathbb{E}_{q(z)}[\log p(x | z)] - \text{KL}(q(z)\|p(z)). \quad (155)$$

Evidence decomposition:

$$\log p(x) = \mathcal{L}(q) + \text{KL}(q(z)\|p(z | x)). \quad (156)$$

11.5 Mean-Field Approximation

Mean-field factorization:

$$q(z) = \prod_{i=1}^M q_i(z_i). \quad (157)$$

Coordinate update:

$$\log q_i^*(z_i) = \mathbb{E}_{q_{-i}}[\log p(x, z)] + \text{const}. \quad (158)$$

11.6 Score-Function Gradient

For $q_\lambda(z)$:

$$\nabla_\lambda \mathcal{L}(\lambda) = \mathbb{E}_{q_\lambda(z)}[(\log p(x, z) - \log q_\lambda(z)) \nabla_\lambda \log q_\lambda(z)]. \quad (159)$$

11.7 Reparameterization Gradient

If $z = g(\lambda, \epsilon)$ with $\epsilon \sim p(\epsilon)$:

$$\mathbb{E}_{q_\lambda(z)}[f(z)] = \mathbb{E}_\epsilon[f(g(\lambda, \epsilon))], \quad (160)$$

$$\nabla_\lambda \mathbb{E}_{q_\lambda(z)}[f(z)] = \mathbb{E}_\epsilon[\nabla_z f(z)|_{z=g(\lambda, \epsilon)} \nabla_\lambda g(\lambda, \epsilon)]. \quad (161)$$

Gaussian reparameterization:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (162)$$

12 Neural Networks as Distribution Parameterizers

12.1 Mixture Density Networks

Conditional mixture model:

$$p_\theta(y | x) = \sum_{k=1}^K \pi_k(x) p_k(y; \theta_k(x)), \quad (163)$$

$$\sum_{k=1}^K \pi_k(x) = 1, \quad \pi_k(x) \geq 0. \quad (164)$$

12.2 Autoregressive Factorization

$$p(x_1, \dots, x_D) = \prod_{d=1}^D p(x_d | x_{<d}). \quad (165)$$

AR(2):

$$x_t = a_1 x_{t-1} + a_2 x_{t-2} + \varepsilon_t, \quad (166)$$

$$\varepsilon_t \sim \mathcal{N}(0, \sigma^2). \quad (167)$$

Neural autoregressive form:

$$x_t | x_{t-1}, x_{t-2} \sim \mathcal{N}(\mu_\theta(x_{t-1}, x_{t-2}), \sigma_\theta(x_{t-1}, x_{t-2})^2). \quad (168)$$

12.3 Gaussian Mean-Variance Head

$$y | x \sim \mathcal{N}(\mu_\theta(x), \sigma_\theta(x)^2). \quad (169)$$

Per-sample Gaussian NLL:

$$\ell(x, y) = \log \sigma_\theta(x) + \frac{1}{2} \left(\frac{y - \mu_\theta(x)}{\sigma_\theta(x)} \right)^2. \quad (170)$$

For fixed residual $r = y - \mu(x)$:

$$\ell(\sigma) = \log \sigma + \frac{r^2}{2\sigma^2}, \quad (171)$$

$$\frac{d\ell}{d\sigma} = 0 \Rightarrow \sigma^2 = r^2. \quad (172)$$

13 Bayesian Neural Networks

13.1 Posterior and Predictive

$$p(w | \mathcal{D}) \propto p(\mathcal{D} | w) p(w). \quad (173)$$

Posterior predictive:

$$p(y^* | x^*, \mathcal{D}) = \int p(y^* | x^*, w) p(w | \mathcal{D}) dw. \quad (174)$$

Monte Carlo approximation:

$$p(y^* | x^*, \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^S p(y^* | x^*, w^{(s)}), \quad w^{(s)} \sim p(w | \mathcal{D}). \quad (175)$$

13.2 Heteroscedastic Regression Loss

If the network outputs $\hat{\mu}_w(x)$ and $s(x) = \log \hat{\sigma}_w^2(x)$:

$$\mathcal{L}_{\text{het}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{(y_i - \hat{\mu}_w(x_i))^2}{2e^{s_i}} + \frac{s_i}{2} \right]. \quad (176)$$

13.3 Mean-Field Variational Family

$$q_\phi(w) = \prod_j \mathcal{N}(w_j; \mu_j, \sigma_j^2). \quad (177)$$

BNN ELBO:

$$\text{ELBO}(\phi) = \mathbb{E}_{q_\phi(w)}[\log p(\mathcal{D} | w)] - \text{KL}(q_\phi(w)\|p(w)). \quad (178)$$

If $p(w_j) = \mathcal{N}(0, \tau^2)$ and $q(w_j) = \mathcal{N}(\mu_j, \sigma_j^2)$:

$$\text{KL}(q_\phi\|p) = \frac{1}{2} \sum_j \left[\frac{\mu_j^2 + \sigma_j^2}{\tau^2} - 1 - \log \frac{\sigma_j^2}{\tau^2} \right]. \quad (179)$$

13.4 Bayes by Backprop

$$w_j = \mu_j + \sigma_j \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, 1). \quad (180)$$

Mini-batch stochastic ELBO:

$$\widehat{\text{ELBO}} = \frac{N}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \log p(y_i | x_i, w) - \text{KL}(q_\phi(w) \| p(w)). \quad (181)$$

13.5 SGLD

$$w_{t+1} = w_t - \frac{\epsilon_t}{2} \nabla_w \tilde{U}(w_t) + \eta_t, \quad (182)$$

$$\eta_t \sim \mathcal{N}(0, \epsilon_t I), \quad (183)$$

where

$$\tilde{U}(w) = -\frac{N}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \log p(y_i | x_i, w) - \log p(w). \quad (184)$$

13.6 MC Dropout

$$\hat{y}^{(t)} = f_{\hat{w}, m^{(t)}}(x^*), \quad t = 1, \dots, T. \quad (185)$$

Predictive mean and epistemic variance:

$$\hat{\mu}(x^*) = \frac{1}{T} \sum_{t=1}^T \hat{y}^{(t)}, \quad (186)$$

$$\hat{\sigma}_{\text{ep}}^2(x^*) = \frac{1}{T} \sum_{t=1}^T (\hat{y}^{(t)} - \hat{\mu}(x^*))^2. \quad (187)$$

If aleatoric variances $\hat{\sigma}_{\text{al}}^{2,(t)}$ are also predicted:

$$\hat{\sigma}_{\text{total}}^2 = \hat{\sigma}_{\text{ep}}^2 + \frac{1}{T} \sum_{t=1}^T \hat{\sigma}_{\text{al}}^{2,(t)}. \quad (188)$$

14 Variational Autoencoders

14.1 Generative Model and Encoder

$$p_\theta(x, z) = p(z) p_\theta(x | z). \quad (189)$$

Encoder:

$$q_\phi(z | x) = \mathcal{N}(z; \mu_\phi(x), \text{diag}(\sigma_\phi(x)^2)). \quad (190)$$

14.2 VAE ELBO

For one sample x :

$$\mathcal{L}_{\theta, \phi}(x) = \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x, z) - \log q_\phi(z | x)] \quad (191)$$

$$= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x | z)] - \text{KL}(q_\phi(z | x) \| p(z)). \quad (192)$$

Dataset objective:

$$\max_{\theta, \phi} \sum_{n=1}^N \mathcal{L}_{\theta, \phi}(x^{(n)}). \quad (193)$$

14.3 Gaussian Reparameterization

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad (194)$$

$$\epsilon \sim \mathcal{N}(0, I). \quad (195)$$

14.4 KL to Standard Normal Prior

If $p(z) = \mathcal{N}(0, I)$ and $q_\phi(z | x) = \mathcal{N}(\mu, \text{diag}(\sigma^2))$:

$$\text{KL}(q_\phi(z | x) \| p(z)) = \frac{1}{2} \sum_{k=1}^d (\mu_k^2 + \sigma_k^2 - \log \sigma_k^2 - 1). \quad (196)$$

15 Proper Scoring Rules, Calibration, and Conformal Prediction

15.1 Proper Scoring Rule

A scoring rule $S(Q, y)$ is proper if, for the true distribution P ,

$$\mathbb{E}_{Y \sim P}[S(P, Y)] \leq \mathbb{E}_{Y \sim P}[S(Q, Y)] \quad (197)$$

for all predictive distributions Q .

15.2 Common Scores

Negative log-likelihood:

$$\text{NLL} = -\log p_\theta(y | x). \quad (198)$$

Binary cross-entropy:

$$\text{NLL}_{\text{bin}} = -[y \log \hat{p} + (1 - y) \log(1 - \hat{p})]. \quad (199)$$

Brier score:

$$\text{Brier} = (\hat{p} - y)^2. \quad (200)$$

CRPS:

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - \mathbf{1}\{y \leq z\})^2 dz. \quad (201)$$

15.3 Calibration and ECE

Perfect binary calibration:

$$\mathbb{P}(Y = 1 | \hat{p} = p) = p. \quad (202)$$

For bins B_b :

$$\text{acc}(B_b) = \frac{1}{|B_b|} \sum_{i \in B_b} \mathbf{1}(\hat{y}_i = y_i), \quad (203)$$

$$\text{conf}(B_b) = \frac{1}{|B_b|} \sum_{i \in B_b} \hat{p}_i. \quad (204)$$

Expected Calibration Error:

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|. \quad (205)$$

15.4 Post-hoc Calibration

Platt scaling:

$$q = \sigma(az + b). \quad (206)$$

Temperature scaling:

$$q = \text{softmax}(z/T), \quad T > 0. \quad (207)$$

15.5 Split Conformal Prediction

Calibration scores:

$$s_i = S(x_i, y_i; \hat{f}), \quad i = 1, \dots, n_{\text{cal}}. \quad (208)$$

Calibration quantile:

$$\hat{q} = \text{Quantile}_{1-\alpha}(\{s_i\}). \quad (209)$$

Prediction set:

$$C(x) = \{y : S(x, y; \hat{f}) \leq \hat{q}\}. \quad (210)$$

Coverage under exchangeability:

$$\mathbb{P}(Y \in C(X)) \geq 1 - \alpha. \quad (211)$$

For regression with residual score $S(x, y) = |y - \hat{y}(x)|$:

$$C(x) = [\hat{y}(x) - \hat{q}, \hat{y}(x) + \hat{q}]. \quad (212)$$

16 Classical Sensor Fusion and Data Association

16.1 Precision-Weighted Gaussian Fusion

For two scalar Gaussian estimates:

$$\hat{x} = \frac{\lambda_1 z_1 + \lambda_2 z_2}{\lambda_1 + \lambda_2}, \quad \lambda_i = \sigma_i^{-2}, \quad (213)$$

$$\sigma_f^{-2} = \lambda_1 + \lambda_2. \quad (214)$$

For N sensors:

$$\sigma_f^{-2} = \sum_{i=1}^N \sigma_i^{-2}, \quad (215)$$

$$\hat{x} = \sigma_f^2 \sum_{i=1}^N \frac{z_i}{\sigma_i^2}. \quad (216)$$

If all sensors have variance σ^2 :

$$\sigma_f = \frac{\sigma}{\sqrt{N}}. \quad (217)$$

16.2 Multivariate Gaussian Fusion

If $\mathbf{y}_i = \mathbf{H}_i \mathbf{z} + \mathbf{r}_i$, $\mathbf{r}_i \sim \mathcal{N}(0, \mathbf{R}_i)$ independently, then

$$\Sigma_f^{-1} = \sum_i \mathbf{H}_i^\top \mathbf{R}_i^{-1} \mathbf{H}_i, \quad (218)$$

$$\mu_f = \Sigma_f \sum_i \mathbf{H}_i^\top \mathbf{R}_i^{-1} \mathbf{y}_i. \quad (219)$$

If $\mathbf{H}_i = I$:

$$\Sigma_f^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1}, \quad (220)$$

$$\mu_f = \Sigma_f (\Sigma_1^{-1} \mathbf{y}_1 + \Sigma_2^{-1} \mathbf{y}_2). \quad (221)$$

16.3 Correlated Sensor Noise

For

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \Sigma_{12} \\ \Sigma_{12} & \sigma_2^2 \end{bmatrix}, \quad (222)$$

the correct scalar fused variance is

$$\sigma_f^2 = \frac{\sigma_1^2 \sigma_2^2 - \Sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\Sigma_{12}}. \quad (223)$$

16.4 Covariance Intersection

For two estimates (μ_1, Σ_1) and (μ_2, Σ_2) :

$$\Sigma_f^{-1} = \omega \Sigma_1^{-1} + (1 - \omega) \Sigma_2^{-1}, \quad (224)$$

$$\Sigma_f^{-1} \mu_f = \omega \Sigma_1^{-1} \mu_1 + (1 - \omega) \Sigma_2^{-1} \mu_2, \quad (225)$$

with $\omega \in [0, 1]$.

16.5 Mahalanobis Distance and Gating

Innovation:

$$v_{ij} = y_j - \hat{y}_i. \quad (226)$$

Mahalanobis distance:

$$D_{ij}^2 = v_{ij}^\top S_i^{-1} v_{ij}. \quad (227)$$

Validation gate:

$$v^\top S^{-1} v \leq \gamma. \quad (228)$$

16.6 Global Nearest Neighbour

Cost matrix:

$$C_{ij} = D_{ij}^2. \quad (229)$$

Assignment problem:

$$\min_{\Theta \in \{0,1\}^{T \times M}} \sum_{i,j} C_{ij} \Theta_{ij} \quad (230)$$

subject to

$$\sum_j \Theta_{ij} \leq 1, \quad \sum_i \Theta_{ij} \leq 1. \quad (231)$$

16.7 PDAF

Observation likelihood:

$$\mathcal{L}_j = \mathcal{N}(y_j; \hat{y}_t, S_t). \quad (232)$$

Association probabilities:

$$\beta_j = \frac{P_D \mathcal{L}_j}{\lambda \bar{c} + P_D \sum_{j'=1}^m \mathcal{L}_{j'}}, \quad (233)$$

$$\beta_0 = \frac{\lambda \bar{c}}{\lambda \bar{c} + P_D \sum_{j'=1}^m \mathcal{L}_{j'}}. \quad (234)$$

Combined innovation:

$$\bar{v} = \sum_{j=1}^m \beta_j v_j. \quad (235)$$

Mean update:

$$\hat{\mu}_{t|t} = \hat{\mu}_{t|t-1} + K_t \bar{v}. \quad (236)$$

Covariance update:

$$\Sigma_{t|t} = \beta_0 \Sigma_{t|t-1} + (1 - \beta_0) \Sigma_{t|t}^* \quad (237)$$

$$+ K_t \left[\sum_{j=1}^m \beta_j v_j v_j^\top - \bar{v} \bar{v}^\top \right] K_t^\top, \quad (238)$$

where

$$\Sigma_{t|t}^* = (I - K_t H_t) \Sigma_{t|t-1}. \quad (239)$$

16.8 JPDA

Marginal association probability:

$$\beta_{ij} = \sum_{\Theta: \Theta_{ij}=1} P(\Theta | \mathcal{Y}_t). \quad (240)$$

Per-track update:

$$\bar{v}_i = \sum_j \beta_{ij} v_{ij}, \quad (241)$$

$$\mu_{i,t|t} = \mu_{i,t|t-1} + K_i \bar{v}_i. \quad (242)$$

17 Deep Multimodal Fusion and Representation Learning

17.1 Embeddings and Cosine Similarity

$$e = f_\theta(X) \in \mathbb{R}^d. \quad (243)$$

Cosine similarity:

$$\text{Sim}(a, b) = \frac{a^\top b}{\|a\| \|b\|}. \quad (244)$$

Cross-modal matching score:

$$\text{score}_{ij} = \frac{(e_{\text{cam}}^{(i)})^\top e_{\text{rad}}^{(j)}}{\|e_{\text{cam}}^{(i)}\| \|e_{\text{rad}}^{(j)}\|}. \quad (245)$$

17.2 Contrastive Learning and InfoNCE

Simple contrastive loss:

$$\mathcal{L}_{\text{contrastive}} = \begin{cases} 1 - \cos(e_{\text{cam}}, e_{\text{rad}}), & \text{same object,} \\ \max(0, \cos(e_{\text{cam}}, e_{\text{rad}}) - m_c), & \text{different object.} \end{cases} \quad (246)$$

InfoNCE:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp((e_{\text{cam}}^{(i)})^\top e_{\text{rad}}^{(i)} / \tau)}{\sum_{j=1}^B \exp((e_{\text{cam}}^{(i)})^\top e_{\text{rad}}^{(j)} / \tau)}. \quad (247)$$

Row-softmax probabilities:

$$p_{ij} = \frac{\exp(A_{ij})}{\sum_{k=1}^B \exp(A_{ik})}, \quad A_{ij} = \frac{(e_{\text{cam}}^{(i)})^\top e_{\text{rad}}^{(j)}}{\tau}. \quad (248)$$

17.3 Temporal Embeddings

$$e_t = \phi_{\text{fuse}}(e_t^{\text{cam}}, e_t^{\text{rad}}, z_t, c_t). \quad (249)$$

Recurrent memory:

$$u_t = [e_t; \Delta z_t; \Delta t; c_t], \quad (250)$$

$$h_t = \text{GRU}(u_t, h_{t-1}). \quad (251)$$

17.4 Fusion Operators

Concatenation plus MLP:

$$e^{\text{fuse}} = \text{MLP}([e^{\text{cam}}; e^{\text{rad}}; e^{\text{lid}}]). \quad (252)$$

Weighted-sum fusion:

$$e^{\text{fuse}} = \sum_{m=1}^M w_m e_m, \quad (253)$$

$$\sum_{m=1}^M w_m = 1, \quad w_m \geq 0. \quad (254)$$

Reliability-based weighting:

$$w_m \propto \sigma_m^{-2}. \quad (255)$$

Gated fusion:

$$g = \sigma(W_g[e_{\text{cam}}; e_{\text{rad}}]), \quad (256)$$

$$e^{\text{fuse}} = g \odot e_{\text{cam}} + (1 - g) \odot e_{\text{rad}}. \quad (257)$$

Mixture of experts:

$$e^{\text{fuse}} = \sum_{k=1}^K \alpha_k(\text{context}) \text{Expert}_k(e_1, \dots, e_M). \quad (258)$$

Product of experts:

$$p(x | z^{(1)}, z^{(2)}) \propto p(x | z^{(1)}) p(x | z^{(2)}). \quad (259)$$

For Gaussian experts:

$$\Lambda = \Lambda_1 + \Lambda_2. \quad (260)$$

18 Attention and Transformers

18.1 Scaled Dot-Product Attention

Score between query i and key j :

$$s_{i,j} = \frac{q_i^\top k_j}{\sqrt{d}}. \quad (261)$$

Attention weights (softmax over keys):

$$\alpha_{i,j} = \frac{\exp(s_{i,j})}{\sum_{j'} \exp(s_{i,j'})}. \quad (262)$$

Context (output) vector:

$$c_i = \sum_j \alpha_{i,j} v_j. \quad (263)$$

Matrix form:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V. \quad (264)$$

18.2 Query, Key, Value Projections

From input embedding e :

$$q = e W^Q, \quad k = e W^K, \quad v = e W^V. \quad (265)$$

18.3 Multi-Head Attention

Each head h uses its own projections W_h^Q, W_h^K, W_h^V :

$$\text{head}_h = \text{Attention}(Q W_h^Q, K W_h^K, V W_h^V), \quad (266)$$

$$\text{MHA}(Q, K, V) = [\text{head}_1; \dots; \text{head}_H] W^O. \quad (267)$$

18.4 Transformer Block (Pre-Norm)

$$\mathbf{x}' = \mathbf{x} + \text{SelfAttn}(\text{LN}(\mathbf{x})), \quad (268)$$

$$\mathbf{x}'' = \mathbf{x}' + \text{MLP}(\text{LN}(\mathbf{x}')). \quad (269)$$

18.5 Sinusoidal Positional Encoding

$$p_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad (270)$$

$$p_{t,2i+1} = \cos\left(\frac{t}{10000^{2i/d}}\right). \quad (271)$$

18.6 Temporal Self-Attention

With positional encoding $\mathbf{p}_t$:

$$\mathbf{q}_t = (\mathbf{e}_t + \mathbf{p}_t)W^Q, \quad \mathbf{k}_\tau = (\mathbf{e}_\tau + \mathbf{p}_\tau)W^K, \quad (272)$$

$$\alpha_{t,\tau} = \frac{\exp(\mathbf{q}_t^\top \mathbf{k}_\tau / \sqrt{d})}{\sum_{\tau'} \exp(\mathbf{q}_t^\top \mathbf{k}_{\tau'} / \sqrt{d})}, \quad (273)$$

$$\hat{\mathbf{e}}_t = \sum_{\tau} \alpha_{t,\tau} \mathbf{v}_\tau. \quad (274)$$

Causal masking: set $\alpha_{t,\tau} = 0$ for $\tau > t$.

18.7 Gaussian Embeddings: Score Statistics

If $\mathbf{q} \sim \mathcal{N}(\boldsymbol{\mu}_q, \Sigma_q)$, $\mathbf{k}_j \sim \mathcal{N}(\boldsymbol{\mu}_{k_j}, \Sigma_{k_j})$ independently:

$$\mathbb{E}[s_j] = \frac{\boldsymbol{\mu}_q^\top \boldsymbol{\mu}_{k_j}}{\sqrt{d}}, \quad (275)$$

$$\text{Var}(s_j) = \frac{1}{d} \left(\boldsymbol{\mu}_q^\top \Sigma_{k_j} \boldsymbol{\mu}_q + \boldsymbol{\mu}_{k_j}^\top \Sigma_q \boldsymbol{\mu}_{k_j} + \text{tr}(\Sigma_q \Sigma_{k_j}) \right). \quad (276)$$

18.8 Uncertainty-Aware Attention

Adaptive temperature:

$$\alpha_j \propto \exp\left(\frac{\mathbb{E}[s_j]}{\sqrt{\text{Var}(s_j) + \epsilon}}\right). \quad (277)$$

Monte Carlo Bayesian attention:

$$\mathbb{E}[\alpha_j] \approx \frac{1}{S} \sum_{s=1}^S \text{softmax}\left(\frac{\mathbf{q}^{(s)\top} \mathbf{k}_j^{(s)}}{\sqrt{d}}\right). \quad (278)$$

18.9 Structured Attention and Sinkhorn

Standard (row-softmax) attention:

$$\alpha_{i,j} = \frac{\exp(s_{i,j})}{\sum_{j'} \exp(s_{i,j'})}. \quad (279)$$

Structured attention (constrained optimization):

$$A^* = \arg \max_{A \in \mathcal{C}} \sum_{i,j} A_{i,j} s_{i,j} - \Omega(A). \quad (280)$$

Sinkhorn normalization (log-space): given logit matrix L , temperature τ , set $P = L/\tau$, then iterate:

$$P_{ij} \leftarrow P_{ij} - \log \sum_j \exp(P_{ij}), \quad (281)$$

$$P_{ij} \leftarrow P_{ij} - \log \sum_i \exp(P_{ij}), \quad (282)$$

$$A = \exp(P) \quad (\text{doubly stochastic}). \quad (283)$$

18.10 Association Logits

Geometric (Mahalanobis-based):

$$\ell_{ij}^{\text{geom}} = -\frac{1}{2}(\mathbf{y}_j - H\mathbf{z}_i)^\top S_i^{-1}(\mathbf{y}_j - H\mathbf{z}_i). \quad (284)$$

Learned (embedding-based):

$$\ell_{ij}^{\text{learn}} = \phi(\mathbf{z}_i)^\top \psi(\mathbf{y}_j). \quad (285)$$

Combined: $\ell_{ij} = \ell_{ij}^{\text{geom}} + \ell_{ij}^{\text{learn}}$.

19 Diffusion Models

19.1 Forward Process

One step:

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, I). \quad (286)$$

Define $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. Closed-form marginal:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I). \quad (287)$$

Transition distribution:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t I\right), \quad (288)$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) I\right). \quad (289)$$

19.2 DDPM Training Objective

Simple (noise-prediction) loss:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} [\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|_2^2]. \quad (290)$$

19.3 Variational Lower Bound

$$\begin{aligned} \mathcal{L}_{\text{VLB}} &= D_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_0) \| p(\mathbf{x}_T)) \\ &+ \sum_{t=2}^T D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)) \\ &- \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1). \end{aligned} \quad (291)$$

19.4 Reverse Sampling

Learned reverse:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \Sigma_t). \quad (292)$$

Ancestral update ($\mathbf{z} \sim \mathcal{N}(0, I)$):

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}. \quad (293)$$

19.5 Score-Noise Relationship

$$\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \propto -\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t). \quad (294)$$

19.6 VP-SDE (Continuous-Time)

Forward:

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x} dt + \sqrt{\beta(t)} d\mathbf{w}_t. \quad (295)$$

Reverse:

$$d\mathbf{x} = \left[-\frac{1}{2}\beta(t)\mathbf{x} - \beta(t)\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt + \sqrt{\beta(t)} d\bar{\mathbf{w}}_t. \quad (296)$$

19.7 Conditional Diffusion and Guidance

Conditional denoiser: $\epsilon_{\theta}(\mathbf{x}_t, t, c)$.

Classifier guidance:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | c) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(c | \mathbf{x}_t). \quad (297)$$

Classifier-free guidance (CFG):

$$\epsilon_{\text{cfg}}(\mathbf{x}_t, t, c) = \epsilon_{\theta}(\mathbf{x}_t, t, \varnothing) + w \left(\epsilon_{\theta}(\mathbf{x}_t, t, c) - \epsilon_{\theta}(\mathbf{x}_t, t, \varnothing) \right). \quad (298)$$

19.8 Diffusion-Based UQ

Given N samples $\mathbf{x}_0^{(1)}, \dots, \mathbf{x}_0^{(N)} \sim p_{\theta}(\mathbf{x} | c)$:

$$\hat{\mathbf{x}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_0^{(n)}, \quad (299)$$

$$\hat{\sigma}^2 = \frac{1}{N-1} \sum_{n=1}^N (\mathbf{x}_0^{(n)} - \hat{\mathbf{x}})^2. \quad (300)$$

Prediction intervals: sort samples, take $\alpha/2$ and $1 - \alpha/2$ quantiles.

20 Bayesian Decision Theory

20.1 Policy, Risk, Optimal Action

Policy (decision rule):

$$\delta : \mathcal{X} \rightarrow \mathcal{A}, \quad a = \delta(x). \quad (301)$$

Risk:

$$R(\delta) = \mathbb{E}[\ell(h, \delta(X))]. \quad (302)$$

Frequentist risk (h fixed):

$$r(\delta | h) = \mathbb{E}_{p(x|h)}[\ell(h, \delta(x))]. \quad (303)$$

Bayesian posterior risk (x observed):

$$\rho(a | x) = \mathbb{E}_{p(h|x)}[\ell(h, a)]. \quad (304)$$

Optimal Bayes action:

$$\delta^*(x) = \arg \min_{a \in \mathcal{A}} \rho(a | x). \quad (305)$$

20.2 Classification (0–1 Loss)

$$\ell_{01}(y^*, \hat{y}) = \mathbf{1}(y^* \neq \hat{y}). \quad (306)$$

$$\delta(x) = \arg \max_y p(y | x) \quad (\text{MAP}). \quad (307)$$

20.3 Regression (ℓ_2 Loss)

$$\ell_2(h, y) = (h - y)^2. \quad (308)$$

$$\delta(x) = \mathbb{E}[h | x] \quad (\text{MMSE} / \text{posterior mean}). \quad (309)$$

20.4 Maximum Expected Utility

$$\delta^*(x) = \arg \max_{a \in \mathcal{A}} \mathbb{E}_{p(h|x)}[U(h, a)]. \quad (310)$$

20.5 Value of Information

MEU with test observation s :

$$\text{MEU}_{\text{test}} = \sum_s p(s) \max_a \mathbb{E}[U(h, a) | s]. \quad (311)$$

Value of perfect information:

$$\text{VPI} = \text{MEU}_{\text{perfect}} - \text{MEU}_{\text{prior}}. \quad (312)$$

21 Multi-Armed Bandits

21.1 Regret

$$\text{Regret}_T = \sum_{t=1}^T [\mu^* - \mu_{a_t}], \quad (313)$$

where $\mu^* = \max_a \mu_a$.

21.2 Upper Confidence Bound (UCB)

Optimistic action score:

$$\tilde{R}_t(a) = \hat{\mu}_t(a) + \sqrt{\frac{2 \ln t}{N_t(a)}}. \quad (314)$$

Action selection:

$$a_t = \arg \max_a \tilde{R}_t(a). \quad (315)$$

21.3 Thompson Sampling

At each step:

$$\tilde{\theta}_t \sim p(\theta | h_t), \quad (316)$$

$$a_t = \arg \max_a R(s_t, a; \tilde{\theta}_t). \quad (317)$$

Posterior update after observing r_t :

$$p(\theta | h_{t+1}) \propto p(r_t | s_t, a_t, \theta) p(\theta | h_t). \quad (318)$$

For Bernoulli bandits with Beta prior:

$$p_a \sim \text{Beta}(\alpha_a + \text{wins}, \beta_a + \text{losses}). \quad (319)$$

21.4 Boltzmann (Softmax) Exploration

$$\pi_{\tau}(a | s) = \frac{\exp(Q(s, a)/\tau)}{\sum_{a'} \exp(Q(s, a')/\tau)}. \quad (320)$$

22 Markov Decision Processes (MDPs)

22.1 MDP Definition

Tuple $\langle \mathcal{S}, \mathcal{A}, p, R, p_0 \rangle$. Transition: $p(s' | s, a)$. Reward: $R(s, a)$.

22.2 Return and Discount Factor

$$G_t = \sum_{k=0}^{T-t-1} \gamma^k r_{t+k} = r_t + \gamma G_{t+1}. \quad (321)$$

22.3 Value Functions

State-value:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right]. \quad (322)$$

Action-value:

$$Q^\pi(s, a) = \mathbb{E}_\pi[G_0 \mid s_0 = s, a_0 = a]. \quad (323)$$

Advantage:

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s). \quad (324)$$

Relationship:

$$V^\pi(s) = \mathbb{E}_{\pi(a|s)}[Q^\pi(s, a)]. \quad (325)$$

22.4 Bellman Optimality Equations

$$V^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} p(s' \mid s, a) V^*(s') \right], \quad (326)$$

$$Q^*(s, a) = R(s, a) + \gamma \sum_{s'} p(s' \mid s, a) \max_{a'} Q^*(s', a'). \quad (327)$$

Optimal policy:

$$\pi^*(s) = \arg \max_a Q^*(s, a). \quad (328)$$

22.5 Value Iteration

$$V_{k+1}(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} p(s' \mid s, a) V_k(s') \right]. \quad (329)$$

Contraction: $\|V_{k+1} - V^*\|_\infty \leq \gamma \|V_k - V^*\|_\infty$.

22.6 Policy Iteration

Evaluation (solve linear system):

$$\mathbf{v} = (\mathbf{I} - \gamma \mathbf{T})^{-1} \mathbf{r}, \quad (330)$$

where $T(s' \mid s) = \sum_a \pi(a \mid s) p(s' \mid s, a)$.

Improvement:

$$\pi'(s) = \arg \max_a [R(s, a) + \gamma \mathbb{E}[V^\pi(s')]]. \quad (331)$$

22.7 POMDP

Observation: $o_t \sim p(o \mid s_t, a_t)$. Belief state:

$$b_t = p(s_t \mid h_t), \quad h_t = (a_1, o_1, \dots, a_{t-1}, o_t). \quad (332)$$

23 Reinforcement Learning

23.1 Monte Carlo RL

Discounted return:

$$G_t = r_t + \gamma r_{t+1} + \dots + \gamma^{T-t-1} r_{T-1}. \quad (333)$$

Update:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \eta (G_t - Q(s_t, a_t)). \quad (334)$$

23.2 Temporal Difference (TD) Learning

TD(0) update:

$$V(s_t) \leftarrow V(s_t) + \eta \underbrace{[r_t + \gamma V(s_{t+1}) - V(s_t)]}_{\text{TD error } \delta_t}. \quad (335)$$

23.3 SARSA (On-Policy TD)

$$Q(s, a) \leftarrow Q(s, a) + \eta [r + \gamma Q(s', a') - Q(s, a)], \quad (336)$$

where $a' \sim \pi(s')$ is the action actually taken.

23.4 Q-Learning (Off-Policy TD)

$$Q(s, a) \leftarrow Q(s, a) + \eta [r + \gamma \max_b Q(s', b) - Q(s, a)]. \quad (337)$$

23.5 Double Q-Learning

$$\begin{aligned} Q_1(s, a) &\leftarrow Q_1(s, a) \\ &+ \eta \left[r + \gamma Q_2 \left(s', \arg \max_{a'} Q_1(s', a') \right) - Q_1(s, a) \right]. \end{aligned} \quad (338)$$

23.6 DQN Loss

$$\mathcal{L}^{\text{DQN}}(w) = \mathbb{E}_{(s, a, r, s') \sim \mathcal{D}} \left[\left(r + \gamma \max_{a'} Q_w(s', a') - Q_w(s, a) \right)^2 \right]. \quad (339)$$

Dueling DQN: $Q(s, a) = V(s) + A(s, a)$.

23.7 Policy Gradient Theorem

$$\nabla_\theta J(\pi_\theta) \propto \mathbb{E}[\nabla_\theta \log \pi_\theta(a \mid s) \cdot Q^{\pi_\theta}(s, a)]. \quad (340)$$

23.8 PPO (Clipped Objective)

$$J^{\text{PPO}} = \mathbb{E} \left[\min \left(\frac{\pi'(a \mid s)}{\pi(a \mid s)} A^\pi, \text{clip} \left(\frac{\pi'(a \mid s)}{\pi(a \mid s)}, 1 - \epsilon, 1 + \epsilon \right) A^\pi \right) \right]. \quad (341)$$

23.9 Deterministic Policy Gradient (DDPG)

$$\nabla_\theta J \approx \mathbb{E} \left[\nabla_\theta \mu_\theta(s) \nabla_a Q_w(s, a) \Big|_{a=\mu_\theta(s)} \right]. \quad (342)$$

23.10 Soft Actor-Critic (SAC)

$$J^{\text{SAC}}(\theta) = \mathbb{E}[R(s, a) + \lambda \mathcal{H}(\pi_\theta(s))]. \quad (343)$$

23.11 ϵ -Greedy Exploration

$$a_t = \begin{cases} \arg \max_a Q(s_t, a) & \text{w.p. } 1 - \epsilon, \\ \text{random action} & \text{w.p. } \epsilon. \end{cases} \quad (344)$$